

Can an AI mathematician be more than just a black box?

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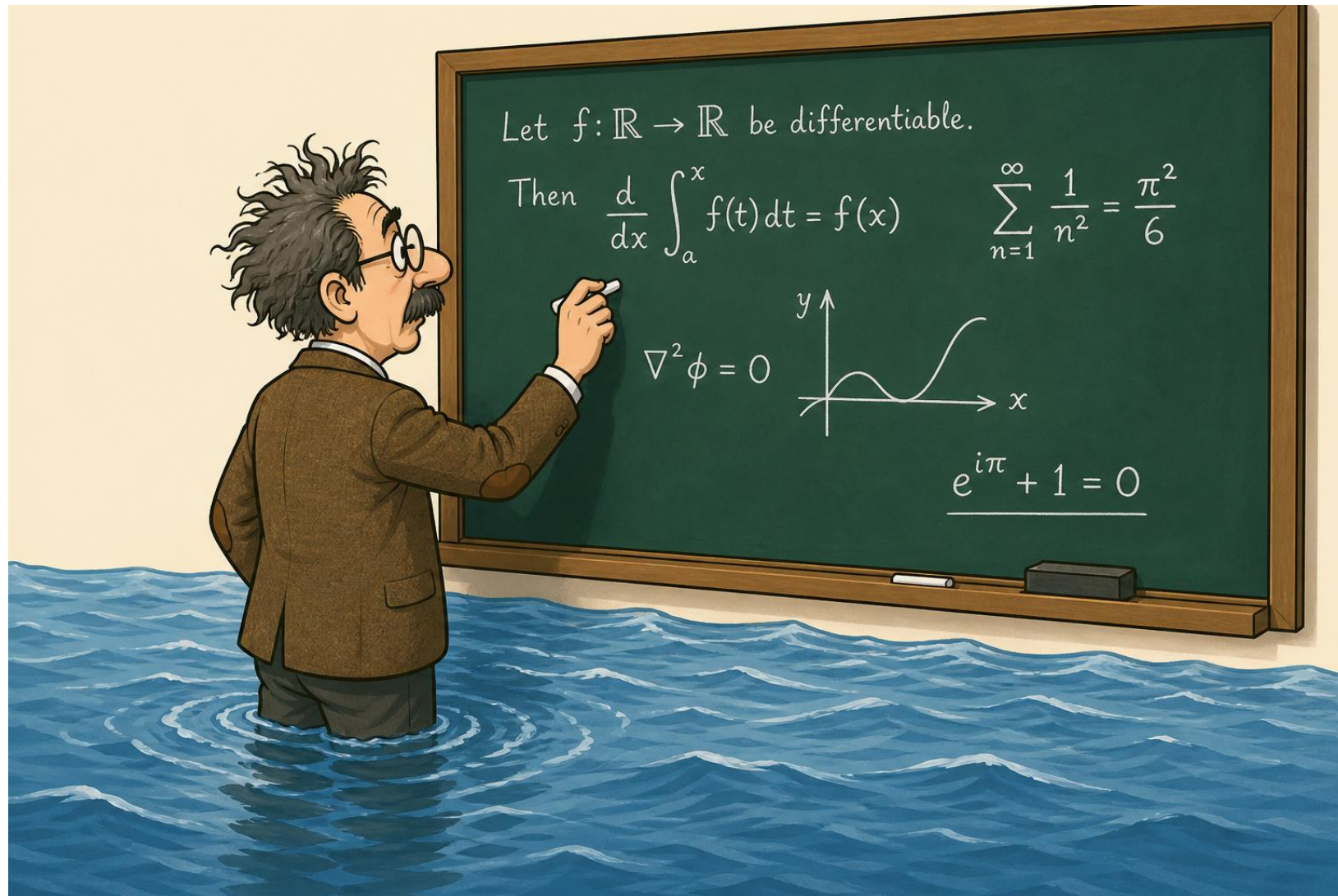
A brief history of LLM capabilities in mathematics.

- (2023) Solved problems at high-school level. Many basic errors and hallucinations.
- (2024) Solved problems at competition level. AlphaProof (a hybrid system) reached silver medal level on the IMO.
- (2025) Several LLMs reached gold medal level on the IMO. Easy research problems started to be solved. Far better reliability. Literature search is completely transformed.
- (2026) LLMs are genuinely useful research partners. They can also autonomously solve problems at a level similar to many PhD students.



What it feels like to be a mathematician today ...

GOSIM Paris 2026



Problem: LLMs work in an opaque way.

- Their chains of thought are not always very informative.
- They often take a somewhat scattergun approach, making lots of guesses until they get lucky.
- When they have a clever idea, it is hard to know how original that idea is.
- We also have very little idea of how much failed search was carried out before the eventual success, and what it looked like.



Why does this matter?

- We want not just a correct proof but *insight*.
- LLMs are not (yet) very good at teaching mathematics.
- A transparent AI mathematician is easier to evaluate.
- If we understand the mathematical thought process better, we should be able to get LLMs to do mathematics *far more efficiently*.



What my group in Cambridge is doing.

- Examining in detail how human mathematicians search for proofs.
- Identifying and automating **proof discovery moves**.
- Building a platform that enables users to generate proofs by **pointing and clicking**.
- Creating a database of **transparent, motivated proofs**.
- Creating new Lean tactics focused on **discovery** rather than **verification**.



Can AI be more than a black box?

Motivated Proof Facilitator **STATEMENT** *A continuous image of a compact set is compact* [← New Statement](#)

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VARIABLES
 X : Topological space
 Y : Topological space
 K : subset of X
 $f : X \rightarrow Y$
 $\mathcal{U}$: family of subsets of Y

HYPOTHESES

- K is compact compact_k
- f is continuous f_continuous
- every element of $\mathcal{U}$ is open in Y u_open_members
- for every $x \in K$, there exists $U \in \mathcal{U}$ such that $f(x) \in U$ u_covers_image

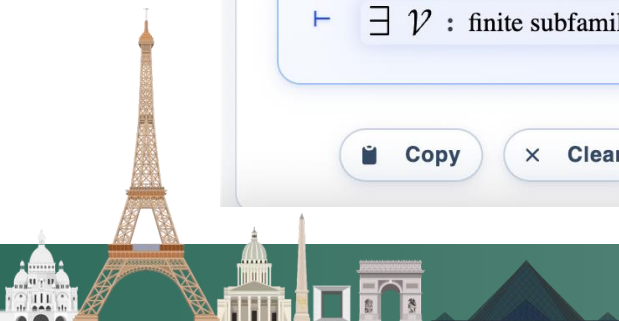
GOALS
 $\vdash \exists \mathcal{V} : \text{finite subfamily of } \mathcal{U} \text{ } \mathcal{V} \text{ covers } f(K)$ image_compact

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Thanks!

